

Standard Treatment Protocols

Clinical Reference — First-Line Management Guidelines

MediAssist Health Network
Clinical Governance Committee
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Access: Doctors and Admin

This document sets out first-line management guidelines for the most common conditions presenting at MediAssist hospitals. Protocols are intended for use by registered medical practitioners and must be applied with clinical judgement. ICD-10 codes are provided to align clinical documentation with billing.

A. Type 2 Diabetes Mellitus

ICD-10: E11

Diagnostic criteria

- Fasting plasma glucose (FPG) ≥ 126 mg/dL, or
- 2-hour plasma glucose ≥ 200 mg/dL on a 75 g OGTT, or
- **HbA1c $\geq 6.5\%$** on a standardised assay.

Pharmacological management

Line	Drug & Dose	Notes
First-line	Metformin 500 mg BD with meals	Titrate to 1000 mg BD over 4 weeks
Second-line	Add Glipizide 5 mg OD	If HbA1c > 7.5% after 3 months on Metformin
Third-line	Consider SGLT2 inhibitor / GLP-1 agonist	Specialist-guided; review renal status

Monitoring

- **HbA1c every 3 months** until stable, then 6-monthly.
- Fasting blood glucose weekly for the first month after any dose change.
- Renal function (eGFR) and urine ACR **annually**; annual fundus and foot screening.

Referral criteria

- HbA1c > 10% despite dual therapy → **endocrinology** referral.
- eGFR < 45 mL/min/1.73m² → **nephrology** referral and Metformin dose review.

B. Hypertension — Stage 2

ICD-10: I10

Diagnostic criteria

Systolic BP ≥ 140 mmHg or diastolic BP ≥ 90 mmHg confirmed on two separate readings at least one week apart, with out-of-office confirmation where feasible.

Pharmacological management

Line	Drug & Dose	Cautions
First-line	Amlodipine 5 mg OD	Titrate to 10 mg if target not met in 4 weeks; ankle oedema
Second-line	Add Telmisartan 40 mg OD	Avoid in pregnancy; monitor K ⁺ and creatinine
Add-on	Hydrochlorothiazide 12.5–25 mg OD	Monitor electrolytes; caution in gout

Monitoring

- BP check at every visit; home BP diary encouraged.
- Serum creatinine and potassium at **baseline and 4 weeks** after ACE-i/ARB initiation.

Lifestyle modification targets

- **Dietary sodium** — less than 2 g/day.
- **Body mass index** — target BMI below 25 kg/m².
- **Alcohol** — restrict to no more than 2 units/day.
- **Aerobic exercise** — at least 30 minutes, 5 days a week.

C. Community-Acquired Pneumonia

ICD-10: J18.9

Severity scoring — CURB-65

One point each for: **C**onfusion, **U**rea > 7 mmol/L, **R**espiratory rate ≥ 30/min, **B**P < 90/60 mmHg, **A**ge ≥ 65 years.

Disposition by total score:

- **Score 0–1** — outpatient management.
- **Score 2** — hospital admission (general ward).
- **Score ≥ 3** — consider ICU / HDU care.

Antimicrobial therapy

Setting	Regimen	Duration
Outpatient	Amoxicillin 500 mg TDS	5 days
Outpatient (atypical)	Add Azithromycin 500 mg OD	3 days
Inpatient	IV Piperacillin-Tazobactam 4.5 g Q8H	Switch to oral when afebrile > 48 h & CRP falling

Monitoring

- SpO₂ monitoring every 4 hours during admission; target 94–98%.
- Repeat chest X-ray at **6 weeks** post-discharge to confirm resolution.

D. Acute Myocardial Infarction — NSTEMI

ICD-10: I21.4

Immediate management (first 60 minutes)

- **Aspirin 300 mg** loading dose stat (chewed).
- **Clopidogrel 300 mg** loading dose (or Ticagrelor per cardiology).
- **IV Heparin 60 units/kg** bolus (maximum 4000 units), then infusion per protocol.
- 12-lead ECG **within 10 minutes** of arrival; high-sensitivity troponin at 0 and 3 hours.

Risk stratification

Calculate the **TIMI** risk score. A score ≥ 3 indicates high risk and warrants an **early invasive strategy (coronary angiography within 24 hours)**.

Ongoing medications

Drug	Dose	Caution
Atorvastatin	80 mg nocte	Check LFTs at baseline
Metoprolol	25 mg BD	Contraindicated in cardiogenic shock / acute LVF
Ramipril	2.5 mg OD from day 2	Only if BP stable; monitor renal function

Escalation

Cardiology on-call must be notified for **all NSTEMI presentations within 15 minutes** of diagnosis.

E. Paediatric Fever Management

ICD-10: R50.9

Temperature-based approach

Temperature	Action
< 38 °C	Observation; encourage fluids
38 – 38.9 °C	Oral Paracetamol; reassess
≥ 39 °C	Paracetamol + tepid sponging
≥ 40 °C or febrile seizure	IV Paracetamol + urgent medical evaluation

Weight-based dosing

Weight	Paracetamol (oral)	Ibuprofen (oral)	IV Paracetamol
< 5 kg	60 mg Q6H	Not recommended	7.5 mg/kg Q6H
5–10 kg	120 mg Q6H	50 mg Q8H	15 mg/kg Q6H
10–20 kg	250 mg Q6H	100 mg Q8H	15 mg/kg Q6H
> 20 kg	500 mg Q6H	200 mg Q8H	15 mg/kg Q6H

Danger signs — immediate escalation

Red flags

Escalate urgently if any of: SpO₂ < 94%, heart rate > 180 in a child < 1 year, altered consciousness, a non-blanching rash, or neck stiffness.

F. Dengue Fever

ICD-10: A90 (dengue); A91 (dengue haemorrhagic fever)

Dengue is endemic across MediAssist's catchment, peaking in the post-monsoon months. Management is supportive; the priority is recognising warning signs and plasma leakage early.

Classification

Category	Features	Disposition
Dengue without warning signs	Fever + 2 of: nausea, rash, aches, leucopenia	Outpatient with daily review
Dengue with warning signs	Abdominal pain, persistent vomiting, mucosal bleed, lethargy, ↑ HCT with ↓ platelets	Admit for IV fluids
Severe dengue	Shock, fluid accumulation with respiratory distress, severe bleeding, organ impairment	ICU management

Management

- **Paracetamol** for fever; **avoid NSAIDs and Aspirin** (bleeding risk).
- Isotonic fluids (0.9% saline / Ringer's lactate) titrated to haematocrit and urine output.
- Monitor **platelet count and haematocrit every 12–24 hours**; watch for a rising HCT with falling platelets (plasma leakage).
- Platelet transfusion only for significant bleeding or platelets $< 10 \times 10^9/L$.

Red flags

Warning signs of plasma leakage warrant immediate admission and IV fluid resuscitation. A narrowing pulse pressure (< 20 mmHg) signals impending shock.

G. Acute Exacerbation of COPD

ICD-10: J44.1

Diagnostic features

Acute worsening of breathlessness, cough and sputum volume/purulence beyond normal day-to-day variation in a known COPD patient.

Management

Intervention	Detail
Controlled oxygen	Target SpO ₂ 88–92%; venturi mask; avoid over-oxygenation
Bronchodilators	Salbutamol 2.5–5 mg + Ipratropium 500 µg nebulised Q6H
Corticosteroid	Prednisolone 40 mg OD oral for 5 days
Antibiotic (if purulent)	Amoxicillin-Clavulanate 625 mg TDS or Doxycycline 100 mg BD
Consider NIV	If pH < 7.35 with hypercapnia despite therapy

Monitoring & escalation

- Arterial blood gas on arrival and after 1 hour of oxygen therapy.
- Escalate to ICU/HDU for NIV failure, worsening acidosis, or reduced consciousness.

Appendix: Key Drug Interactions & Cautions

The interactions below recur across the protocols in this document and warrant specific attention at prescribing.

Combination	Risk	Action
Azithromycin + other QTc-prolonging drugs	Prolonged QTc, arrhythmia	Review ECG; avoid stacking
ACE-i/ARB + Spironolactone	Hyperkalaemia	Monitor potassium closely
Metformin + iodinated contrast	Lactic acidosis if AKI	Withhold around contrast if eGFR low
Warfarin + Metronidazole/Macrolide	↑ INR, bleeding	Monitor INR; dose-adjust
NSAIDs + ACE-i + Diuretic ('triple whammy')	Acute kidney injury	Avoid combination
Clopidogrel + high-dose PPI	↓ antiplatelet effect	Prefer pantoprazole

Disclaimer

This document is a guideline, not a substitute for clinical judgement or the patient-specific summary of product characteristics. Report adverse drug reactions to the Pharmacovigilance desk.

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